

Cross-validation summary

Cross-validation was performed using **5-fold cross-validation** to evaluate the stability and generalization ability of the **Random Forest model**. In this method, the dataset is divided into five equal parts. In each iteration, four parts are used for training and one part is used for testing. This process is repeated five times so that each part of the dataset is used once for testing.

The cross-validation scores obtained for the five folds are : **0.9995, 0.999, 0.655, 0.998, 0.999**

The **average cross-validation accuracy** of the model is **0.9301 (93.01%)**.

Although most folds achieved very high accuracy close to **99%**, one fold produced a lower accuracy (**0.655**). This indicates that the model performs very well on most data splits but may show slightly lower performance on some subsets of the data.

Overall, the Random Forest model still demonstrates **strong predictive performance and good generalization ability**, making it a suitable model for this dataset.